



The UNIVERSITY of OKLAHOMA

# Computer Vision for Autonomous Vehicles - Fall 2025

## Guest Lecture Series



### *Building Scalable Spatial Intelligence: Distributed Optimization and Hierarchical Neural SLAM*

***Dr. Yulun Tian, University of Michigan***

***Thursday Dec 11<sup>th</sup> 2025, 1:30 PM, Carson 438***

**Abstract:** As autonomous systems move toward large-scale networked operation, they must construct reliable and globally consistent “mental models” that scale to large environments and long mission durations. In this talk, I will present two complementary lines of research toward this objective. In the first part, I focus on scaling SLAM to teams of robots operating in GPS-denied environments. I will examine the underlying computational problems that arise in collaborative SLAM and present distributed optimization algorithms that exploit Riemannian geometry and problem sparsity to provide performance guarantees and support robust real-world multi-robot deployment. The second part of the talk shifts to scaling and improving the underlying environment representation. I will present a neural implicit SLAM framework that enables high-fidelity and differentiable 3D scene modeling, and discuss how hierarchical map representations and multiresolution optimization strategies allow such neural approaches to remain efficient and scalable. I will conclude by returning to the broader vision and highlight open opportunities at the intersection of distributed optimization, neural implicit representations, and large-scale spatial intelligence.

**About speaker:** Dr. Yulun Tian is an Assistant Professor of Robotics at the University of Michigan. His research focuses on spatial perception, SLAM, and multi-agent systems. His current work applies tools from AI and machine learning, nonlinear and distributed optimization, and graph theory to develop principled algorithms with theoretical guarantees and real-world systems for multi-agent perception and navigation. His work received the 2024 Best Dissertation Award from the IEEE RAS Technical Committee for Multi-Robot Systems, the 2022 King-Sun Fu Memorial Best Paper Award from the IEEE Transactions on Robotics, a 2021 Honorable Mention from the IEEE Transactions on Robotics, and a 2020 Honorable Mention from the IEEE Robotics and Automation Letters.